CGS410 COURSE PROJECT - REPORT

Project: Structural Efficiency, Cognitive Constraints and Morphological Anchoring in Large Language Models: A Cross-Linguistic Analysis

By : Ankit(240134), Om Appasaheb Nuli (240713) & Chaitanya Sudhir Gaikwad (240292)

PART 1: Emergent Syntactic Efficiency: A Cross-Linguistic Study of Dependency Length Minimization in Grok and Qwen

Motivation for the Research Problem

Background and context

- In most languages, related words stay close together - a pattern called **Dependency Length Minimization (DLM)** - which makes sentences easier for our brains to process.
- Large Language Models (LLMs) like Qwen and Grok, however, are trained simply to predict the next word. They lack fixed grammar rules and have no inherent concept of human cognitive load.
- This raises a key question: *Even with such simple training, do LLMs naturally learn human-friendly patterns like DLM just by processing vast amounts of human-written text?*

Why this problem is important ?

For understanding language and the human mind : DLM and centre-embedding are usually explained by limits of human memory. But if LLMs (which don't have human memory) show the same patterns, it suggests these patterns may naturally come from learning large amounts of language, not just human limitations.
For building better AI : If an LLM ignores DLM, its sentences may be correct but hard to read. If it struggles with complex nested sentences, it may fail in technical or legal writing. Knowing this helps us improve and evaluate AI better.
The gap this project fills : Most studies focus only on human text. This project compares human text, LLM-generated text and random text across six languages. It also tests two different-sized models to see if bigger models handle complex sentences better.

Research Problem

This project addresses the following research question: Do Qwen and Grok produce text that exhibits Dependency Length Minimization patterns comparable to human language across six typologically diverse languages and do Qwen and Grok all demonstrate human-like processing breakdowns for deeply centre-embedded sentences?

Our study has two specific objectives:

- We compare how close related words are (dependency length) in text from Qwen and Grok with real human-written data across six languages (Chinese, English, Indonesian, Spanish, Thai, Vietnamese). This helps us see if DLM naturally appears in LLMs.
- We check how well models like GPT-2 and Qwen understand difficult, nested English sentences. We also see if bigger models handle these complex sentences better than smaller ones.

Hypothesis and Predictions

Hypothesis : Do LLMs naturally write efficient sentences? (DLM)
Null Hypothesis (H_0): LLM word order is effectively random; they do not learn to keep related words close together.
Alternative & Prediction (H_1): Both models will group related words significantly closer together than random chance

(showing lower Mean Dependency Length) across all six languages. This suggests human-like sentence efficiency (DLM) naturally emerges during training.

Methods

Data

Human-authored text was sourced from Universal Dependencies (UD) treebanks (version 2.12) for all six languages. The selection covers a broad typological range:

- Chinese (zh) — SVO, isolating, no inflectional morphology, tonal
- English (en) — SVO, analytic, morphologically sparse
- Indonesian (id) — SVO/flexible, agglutinative, no tonal distinctions, morphologically moderate
- Spanish (es) — SVO with flexible order, fusional morphology, rich verbal inflection
- Thai (th) — SVO, analytic, isolating, tonal, no inflectional morphology
- Vietnamese (vi) — SVO, analytic, isolating, tonal, classifier-based

For each language, all sentences from the train and development splits were used, yielding between 5,000 and 12,000(varying) sentences per language. The inclusion of four Southeast and East Asian languages (Chinese, Indonesian, Thai, Vietnamese) alongside two European languages (English, Spanish) ensures the analysis is not dominated by Indo-European typological properties.

- For the DLM experiment, sentences from two LLMs were used - Qwen and Grok - each processed through the same Stanza-based parsing pipeline as the human data to ensure comparability.

Grok-generated sentences were generated and stored as JSON files at LLM/LLM_sentences_grok/<lang>_llm_sentences.json for each language. These were read and parsed by the script LLM/llm_parser.py, which runs each sentence through Stanza to produce dependency annotations in the same format as the human output. Results were written to LLM/output_grok/<language>_llm.json.

Qwen-generated sentences were produced using a dedicated generation script, LLM_sentence_generation_Qwen.py, and stored in CSV format in the LLM_sentences_Qwen/ folder. The same llm_parser.py pipeline was then applied to parse these sentences through Stanza and compute dependency lengths.
By using the same Stanza parsing pipeline for both human and LLM-generated sentences, the comparison is controlled: any differences in measured dependency length reflect genuine differences in sentence structure rather than artefacts of different parsing tools.

- A random baseline was computed for both human and LLM-generated sentences. For each sentence, words were randomly shuffled and the shuffled sentence was re-parsed through Stanza to measure the resulting dependency length. This process produces a linguistically meaningless upper bound; it represents the expected dependency length if word order carried no structural information. By comparing observed LLM and human dependency lengths to this random baseline, we can assess how much structural efficiency each text source achieves above pure chance.

Dependency Length Measurement

Following Futrell et al. (2015), dependency length for a word w is defined as the absolute difference in linear position between w and its syntactic head : **DL(w) = | Position(w) – Position[head(w)] |**

Mean dependency length (MDL) for a sentence is the average DL over all non-root tokens. The random baseline is computed by permuting word positions 1,000 times per sentence and computing the expected MDL under the uniform permutation distribution.

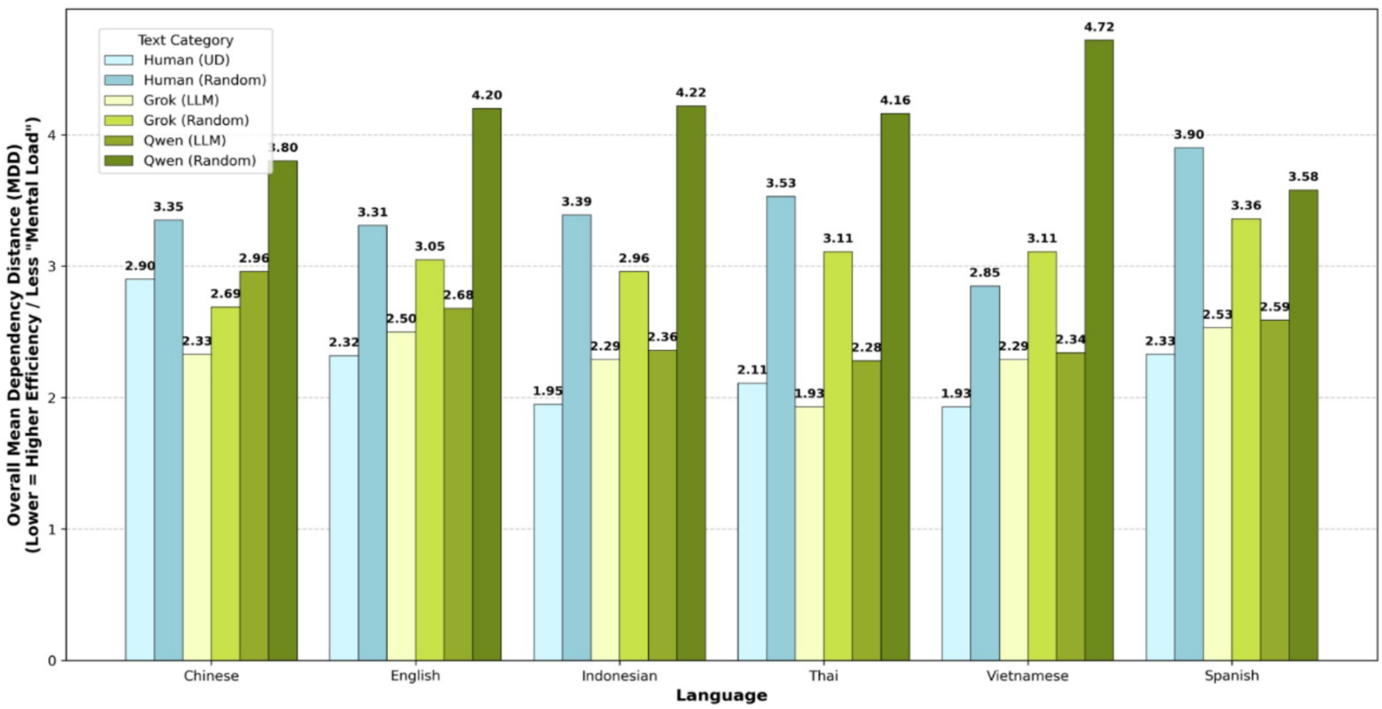
Testing Hypothesis : While beating the random chance baseline confirms the LLM's ability to learn the structure, the hypothesis tests whether LLM is on par with human writers in terms of its abilities. The Mann-Whitney U test allows for comparing the distribution of MDL scores for LLM generated sentences with the distribution of scores for human UD treebanks. If there is no significant difference between those distributions, it means that the sentence structure generated by LLM is not statistically distinguishable from humans for that language.

Results

Mean Dependency Length across all six languages

Languages	Human (UD)	Human (Rand.)	Grok (LLM)	Grok (Rand.)	Qwen (LLM)	Qwen (Rand.)
Chinese	2.90	3.35	2.33	2.69	2.96	3.80
English	2.32	3.31	2.50	3.05	2.68	4.20
Indonesian	1.95	3.39	2.29	2.96	2.36	4.22
Thai	2.11	3.53	1.93	3.11	2.28	4.16
Vietnamese	1.93	2.85	2.29	3.11	2.34	4.72
Spanish	2.33	3.90	2.53	3.36	2.59	3.58

Cross-Linguistic Comparison: Language Efficiency (Human vs. AI vs. Random)



Statistical Results : Mann–Whitney U test results show extremely small p-values ($p \ll 0.05$), confirming that dependency lengths in real language (human/LLMs) are significantly shorter than in random data.
The −log10(p) transformation highlights very high significance across all models (image attached in drive link in appendix).
This provides strong statistical evidence for structured efficiency in language.
Overall, results are robust and consistent across datasets and models.

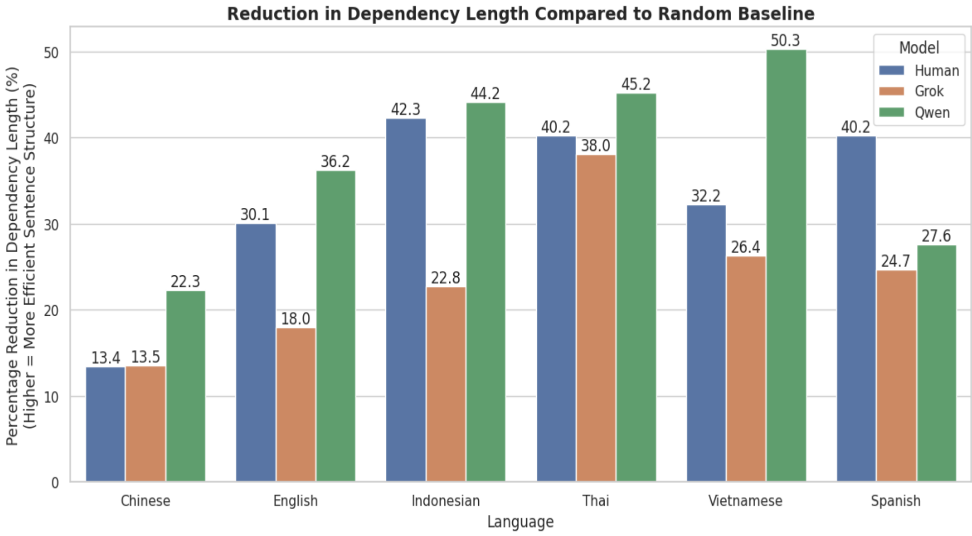
Human p-value	Grok p-value	Qwen p-value
5.347131e-142	2.572388e-193	8.668948e-248
0.000000e+00	0.000000e+00	0.000000e+00
0.000000e+00	0.000000e+00	0.000000e+00
0.000000e+00	0.000000e+00	0.000000e+00
7.386491e-220	0.000000e+00	0.000000e+00
0.000000e+00	0.000000e+00	0.000000e+00

There were considerable decreases in dependency lengths in comparison with random baselines across all six languages under consideration.

Languages	Human Reduction(%)	Grok Reduction(%)	Qwen Reduction(%)
Chinese	13.389717	13.513569	22.278667
English	30.094777	17.984484	36.199684
Indonesian	42.27923	22.780851	44.187631
Thai	40.42853	38.048321	45.242049
Vietnamese	32.199305	26.351023	50.340924
Spanish	40.205831	24.695566	27.634480

Human language consistently achieved the high reduction although Qwen achieved highest, indicating strong optimization for efficient sentence processing. LLMs (Grok and Qwen) also demonstrated significant reduction, though Grok generally lower than human data.

Random baselines exhibited the highest dependency lengths, confirming the absence of structural optimization This pattern was consistent across all languages, suggesting that DLM is a robust cross-linguistic phenomenon.



Theoretical Implications

DLM: What the results tell us about language and learning

- Both Qwen and Grok produce sentences where related words stay closer together than in random text. This shows that DLM naturally appears in these models just by learning from human language.
- There are cases where Grok retains words closer to those used by humans, particularly for languages like Chinese and Thai. The reason is that the model might exaggerate the pattern since it learned from simple sentences.
- Qwen presents high values when making random comparisons because it produces longer sentences. It thus makes the difference between the two models much more significant.
- Differences are also noticeable among the models; Qwen always follows the human pattern, while Grok occasionally exceeds it. Such behaviour can be attributed to disparities in training datasets and methods like RLHF.
- Conclusion: DLM is not an obstacle in human language but a natural pattern that any model can learn from sufficient linguistic data.

The broader perspective

Phenomena that we might attribute to limitations of the human brain, such as DLM or problems with extremely nested sentences, could in fact arise due to how languages are acquired and used. This is evidenced by the presence of the same tendencies in artificial intelligence language models. Therefore, these constraints may not be inherent biological traits but rather the result of acquiring language by humans. In the case of complexly nested sentences, it is possible that these structures do not occur in languages simply due to their difficulty, rather than their impossibility.

PART 2: Centre-Embedding and Working Memory in Language Models: A Comprehension and Attention Analysis of GPT-2 and Qwen2.5

Research Problem

Centre-embedded sentences - e.g. "The rat that the cat that the dog chased killed ate the malt" - require holding partially processed subjects in memory until their corresponding verbs appear. Human comprehension collapses sharply beyond depth two, taken as evidence for a finite working-memory buffer. Transformer LLMs use full-context parallel attention rather than a serial buffer, yet whether this architectural difference confers any practical advantage on centre-embedded structures remains an open empirical question.

Research Question: Do transformer LLMs exhibit a working-memory-like degradation - in comprehension accuracy and attention-based cue retrieval - as centre-embedding depth increases?

Hypothesis 2: Do LLMs hit a working memory wall like humans ? (Centre-Embedding)

Null Hypothesis (H_0) : Adding nesting levels makes no difference. Comprehension accuracy remains constant from depth 1 through 4.

Alternative Hypothesis (H_1) : Both models will show a steady accuracy drop, with a sharp collapse beyond depth 2. This suggests LLMs hit a "functional wall" due to the rarity of deeply nested structures in training data, mimicking human biological limits.

Hypothesis and Objective: We hypothesize that both comprehension accuracy and the attention retrieval gap (how cleanly a verb attends to its correct subject) decrease monotonically with depth 1→4, suggesting a functional analog to human working-memory limits. We test this on GPT-2 (117M, base) and Qwen2.5-0.5B-Instruct (500M, instruction-tuned) to assess whether scale or instruction-tuning provides any advantage.

Predictions

- Accuracy will drop sharply beyond depth 2. No model (regardless of size) is expected to maintain high performance at depths 4 because the issue is structural rarity, not model capacity.
- Attention weights :

- Low depths : Attention focuses correctly on relevant subjects and verbs.
-High depths : Attention becomes confused by multiple similar nouns, leading the model to forget subject-verb pairings, mirroring human interference in working memory.

Methods

Dataset: We procedurally generated 40 sentences per depth (depths 1–4, 160 total) using unique animal names, verbs, and objects sampled without replacement from pools of 30, 24, and 20 items respectively (seed = 0). Each depth-d sentence contains d+1 subject–verb–object triples. Four syntactic templates (that-relative, which-relative, bare, comma-that) prevent reliance on a single complementizer cue. Total comprehension questions: 40×2 + 40×3 + 40×4 + 40×5 = 560.
Comprehension Metric: For each subject–verb–object triple (s_i, v_i, o_i) the model answers "Who did the action to object ?" in up to 8 tokens (greedy decoding).
Accuracy at depth d: $Acc(d) = (1 / N_d) \cdot \Sigma\, 1[\hat{s} = s_i]$, where N_d is the total question count at depth d and $1[\cdot]$ is the indicator function.(It computes accuracy as the fraction of correct answers out of all N_d questions at depth d. The term s_i (inside the indicator) means the model's predicted subject equals the true subject s_i ; it counts as 1 if correct, otherwise 0.)
Attention Retrieval Gap: Attention weights from the last four transformer layers are averaged across heads. For verb token v, target subject s, and distractor nouns $\{d_i\}$: $Gap(d) = A(v \to s) - \max_i A(v \to d_i)$
A positive gap indicates correct subject retrieval; a negative gap indicates distractor interference. Both models run in eager-attention mode on the same hardware.

Results

Depth	GPT-2 Acc.	Qwen Acc.	GPT-2 Gap	Qwen Gap
1	0.50	0.67	+0.0011	+0.0004
2	0.29	0.43	−0.0063	−0.0035
3	0.22	0.22	−0.0101	−0.0050
4	0.13	0.13	−0.0120	−0.0058

Table 1: Accuracy and retrieval gap by model and depth

Both accuracy and retrieval gap decrease monotonically with depth for both models (Table 1). Qwen starts significantly higher at depth 1 (0.67 vs 0.50) but both converge to 0.13 by depth 4, indicating that instruction-tuning helps at shallow depth but fails under severe embedding. The retrieval gap turns negative at depth 2 for both models, meaning verbs attend more to wrong nouns than to the correct subject - a signature of distractor interference that grows monotonically worse.

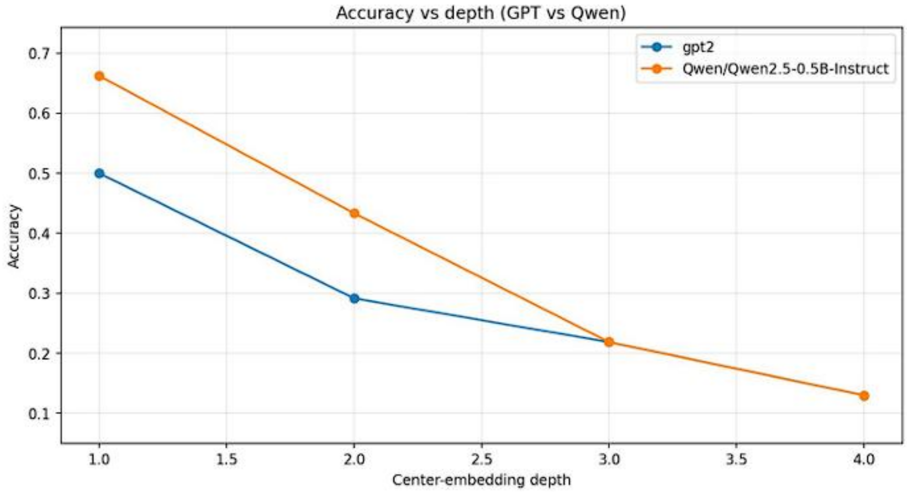


Figure 1: Comprehension accuracy vs. centre-embedding depth (both models).

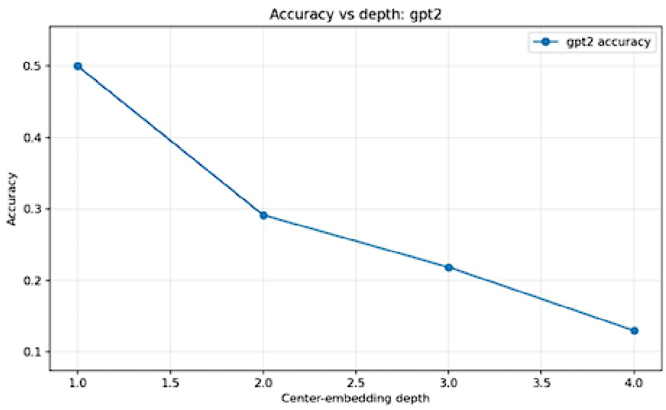


Figure 2: GPT-2 accuracy (0.50→0.13).

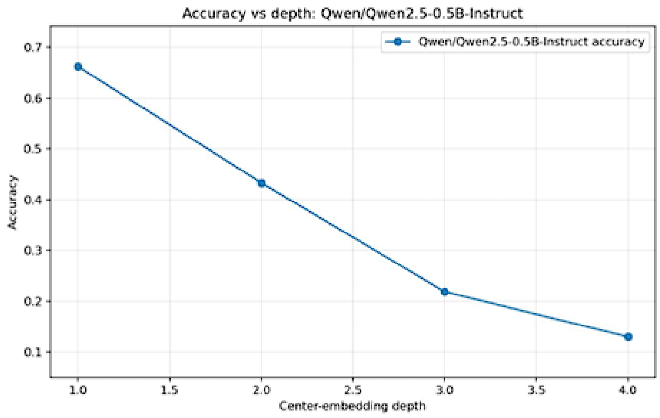


Figure 3: Qwen accuracy (0.67→0.13).

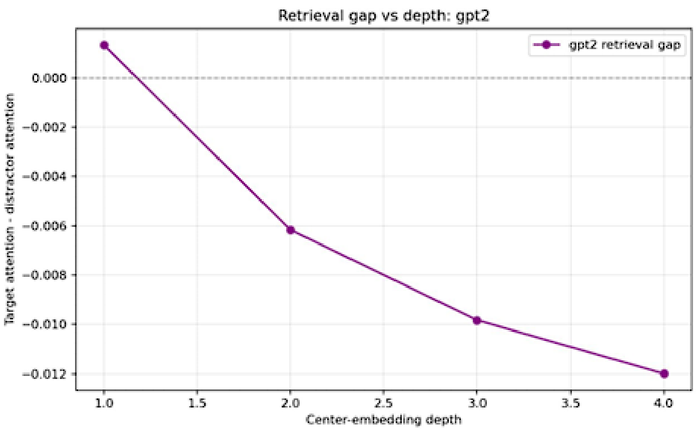


Figure 4: GPT-2 retrieval gap - negative from depth 2

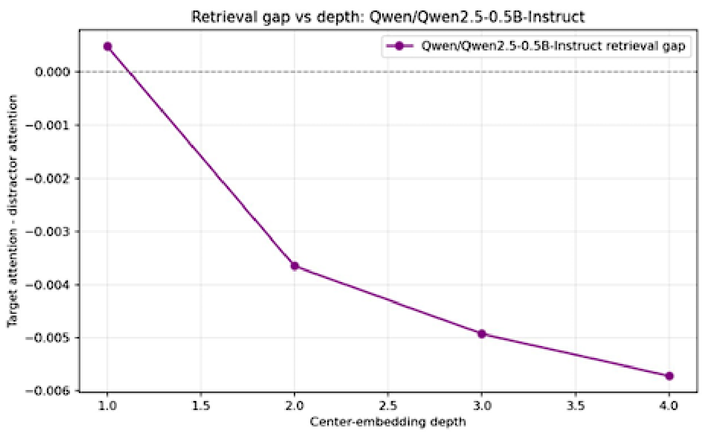


Figure 5: Qwen retrieval gap - mirrors GPT-2

Theoretical Implications

Machine Language Processing

- **Attention Breakdown:** Although LLMs can theoretically process all words at once, they get confused when too many nouns compete. Verbs get distracted by the wrong nouns, mirroring human memory interference.
- **Training vs. Limits:** Better training helps models handle simple sentences (depth 1), but that advantage vanishes at deeper levels (depths 3-4). This shows that while training improves efficiency, it does not fix the core capacity limit.
- **Real-World Impact:** LLMs used for complex writing (like legal documents or nested code) might quietly connect the wrong words.

Language Evolution

- The fact that both humans and AI fail at deeply nested sentences shows that this isn't just a biological flaw in the human brain.
- The universal avoidance of deeply nested sentences (centre-embedding) is a fundamental limit for *any* system trying to process structured language, whether biological or artificial.

Conclusion

- As sentence complexity (depth) increases, model accuracy steadily drops, confirming the hypothesis.
- By depth 4, all models-regardless of their size or how they were trained-fail completely, performing no better than random guessing.
- Ultimately, LLMs share the same "working memory" bottleneck as humans, proving that this limitation is a universal feature of language processing systems.

PART 3 : How Case Marking Helps Languages Handle Long-Distance Word Relationships

Motivation

Sentences are generally easier to process when related words are kept close together, a principle known as **Dependency Length Minimization (DLM)**. Yet, languages like Hindi, Japanese, and German frequently break this rule. They remain comprehensible by using **case markers**—small tags that clearly identify a word's grammatical role (e.g., "who did it").

Project Objective: To prove that case markers act as signposts that neutralize the cognitive penalty of long-distance word connections, making them easier to process.

Hypotheses and Predictions

Hypothesis 1: Case markers flatten the difficulty curve.
Prediction: In languages without case markers (like English), reading difficulty will spike as words get further apart. In languages with case markers, reading difficulty will barely increase (or even decrease) with distance.
Hypothesis 2: Removing markers breaks the sentence.
Prediction: If we artificially delete case markers from sentences, reading difficulty will shoot up, and this spike will be much worse for words that are far apart compared to words that are close.
Hypothesis 3: Uniform Information Density (UID).
Prediction: Because case markers make difficult sentences easier, case-marking languages will have a lower variance in reading difficulty across the sentence. Difficulty will be spread out smoothly.

Methods

Project Analysis Workflow

High-level overview of the computational linguistic pipeline.

Data

We used the Universal Dependencies dataset, examining 10 distinct languages. Five of these languages have case markers (Hindi, Japanese, Korean, Turkish, German) and five do not (English, French, Chinese, Indonesian, Vietnamese). The total dataset includes over 1.8 million word relationships.

Experimental Methods

To measure 'reading difficulty', we used an AI language model (XGLM). The AI reads a sentence word by word. If a word is totally unexpected, the AI gives it a high 'difficulty' score (called surprisal). We calculate Surprisal S for a word w_i as:

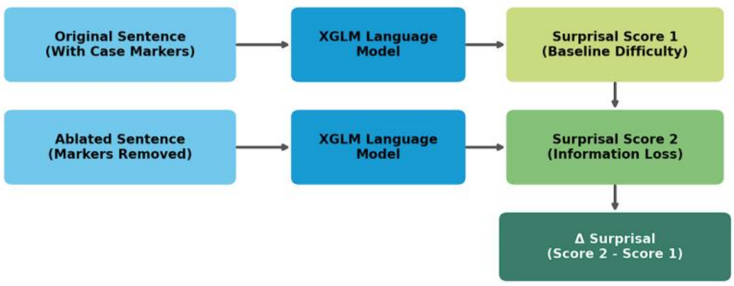
$$S(w_i) = -\log_2 P(w_i \mid w_1, ..., w_{i-1})$$

where $P(w_i \mid w_1, ..., w_{i-1})$ represents the probability of encountering the current word w_i given the context of all preceding words.

For Hypothesis 2, we performed an 'ablation' experiment. We computationally identified and deleted the case markers from the sentences in the datasets. We then fed the broken sentences back into the AI model to get new difficulty scores. We measured information lost (Delta Surprisal) as:

$\Delta S = S(\text{ablated}) - S(\text{original})$

Ablation Experiment Methodology



How We Analyse the Patterns (Inference)

To understand if case markers truly help readers, we don't just look at one or two sentences. We use a statistical 'pattern-finder' (Regression) that looks at millions of word pairs across 10 languages to find the underlying truth. Our logic follows three simple steps:

1. The 'Distance Tax': In almost all languages, the number of words between two related words represents a kind of 'tax' that complicates the comprehension process. In other words, as the number of words grows, Surprisal shoots up.
2. The 'Case Marker Rebate': There's an explicit mathematical relationship that should be observed if our theory is correct: namely, that case markers give us a 'rebate' which neutralizes this negative effect of increasing the distance. With their help, languages such as Hindi have already defined the role of words.
3. Leveling the Playing Field: We control for the confounds (e.g., word frequency, length). To this end, our analysis 'filters out' these factors and measures only those changes to reading difficulty driven by the grammar itself.

$$\text{Difficulty} \sim \text{Distance} + \text{Case\_Marking} + (\text{Distance} \times \text{Case\_Marking}) + \text{Controls}$$

In order to confirm this interaction, we measure the Impact term (Distance × Case_Marking). Last but not least, we carry out the 'stress test' through the ablation technique (deleting case markers).

Results

Hypothesis 1: Slopes and Difficulty Curves

We predicted that case markers flatten the difficulty curve. The results strongly confirm this. Table 1 shows that for non-case-marking languages, difficulty goes up as distance increases (positive slope). For case-marking languages, difficulty actually goes down (negative slope).

Language	Type	Distance Effect (Slope)	p-value
Hindi	Case	-0.223	< 0.001
Japanese	Case	-0.212	< 0.001
Korean	Case	-0.215	< 0.001
Turkish	Case	-0.148	< 0.001
German	Case	+0.022	< 0.001
English	Non-case	+0.063	< 0.001
French	Non-case	+0.064	< 0.001
Indonesian	Non-case	+0.105	< 0.001
Vietnamese	Non-case	+0.064	< 0.001
Chinese	Non-case	+0.022	< 0.001

Our OLS regression model confirmed this statistically. The interaction coefficient (β_3) was highly significant (-0.212, $p < 0.001$), proving that case markers neutralize the distance penalty. Figure 1 visualizes this stark contrast:

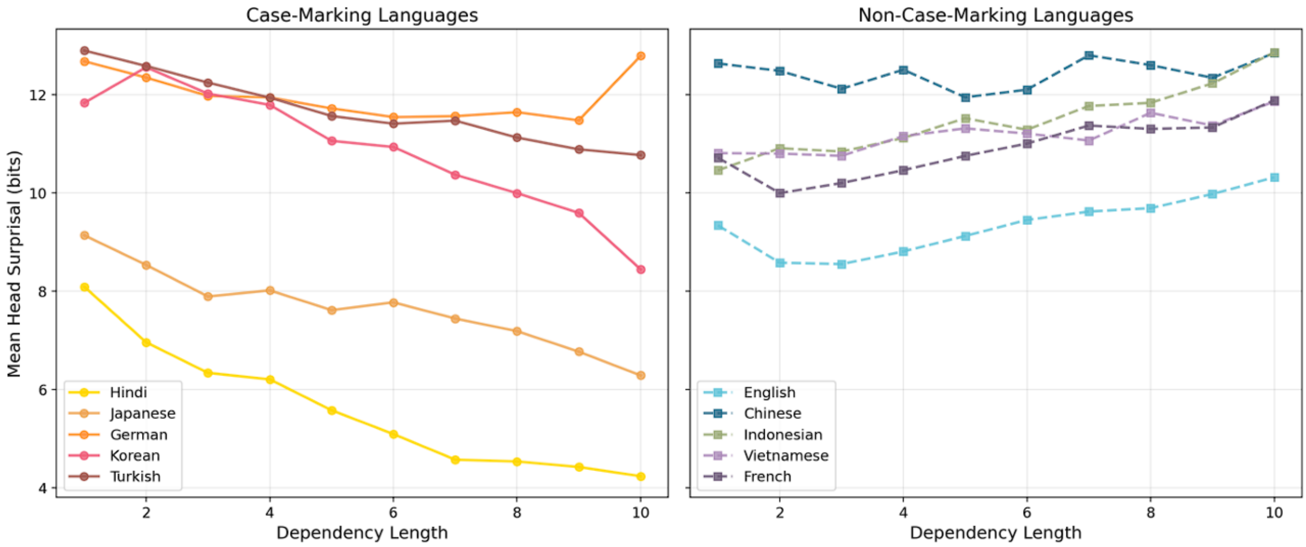


Figure 1: Side-by-side comparison. Case-marking languages (left) vs Non-case-marking languages (right)

Hypothesis 2: Removing Markers (Ablation)
To truly prove that case markers are doing the heavy lifting, we designed an ablation experiment. We used Python scripts to parse the grammatical trees of the sentences and physically delete the case markers. We then asked the AI model to re-evaluate the sentences. We predicted that removing the markers would severely break the sentence's readability. The results showed an average penalty of +3.04 bits of difficulty when markers were removed. Figure 2 shows the exact amount of difficulty added per language when we stripped away their case markers.

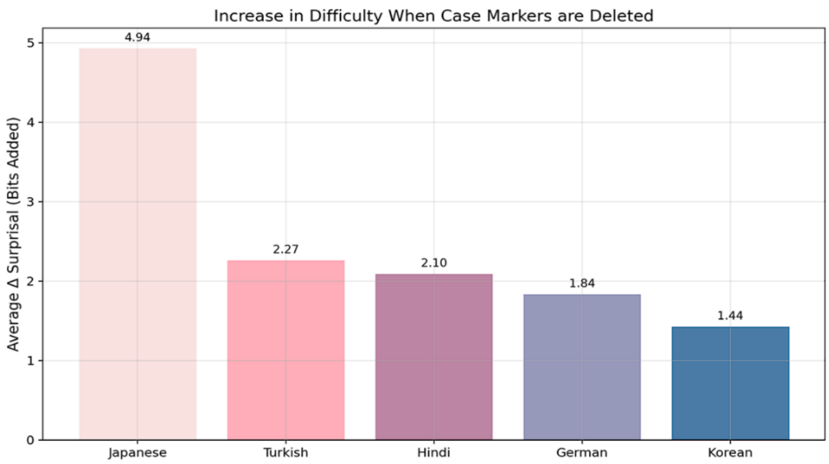


Figure 2: Average increase in reading difficulty (Δ Surprisal) when case markers are deleted.

Hypothesis 3: Uniform Information Density (UID)
We predicted that case markers would smooth out the overall difficulty of the sentence, leading to lower variance in difficulty scores (UID). Interestingly, we found the opposite. The variance (Coefficient of Variation) was not lower for case-marking languages

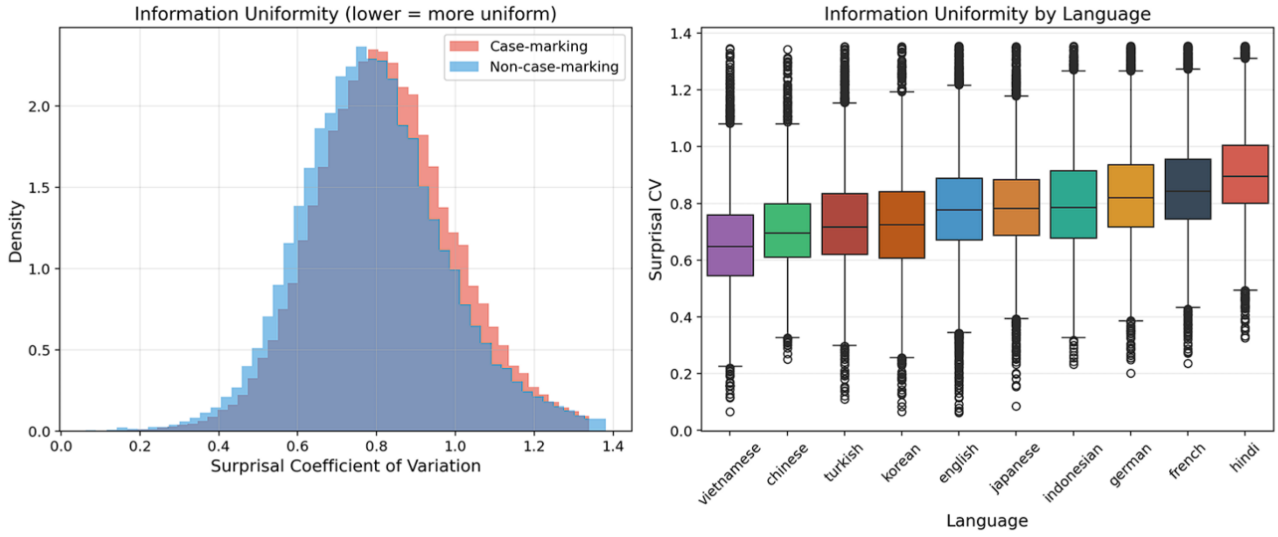


Figure 3: Variance in difficulty. Case-marking languages do not have a smoother distribution.

This surprising result indicates that case markers do not act as global smoothers for the entire sentence. Instead, they act as highly localized anchors - they only reduce difficulty precisely at the moment a long-distance connection is finally resolved, rather than smoothing the entire sentence.

Theoretical Implications

In order to examine the implications of these findings, we outline their usefulness, possibilities for further investigation, and present limitations:

Usefulness to Language Research
• Cognitive Insights: The results of our paper indicate that human working memory makes use of structural "anchors". Case markers are used explicitly as signs which make the working memory more comfortable handling long dependencies. This challenges current psycholinguistic theories which consider long-distance to automatically be difficult to process.
• AI Analysis: This paper should be seen as an alarm for analyzing Large Language Models (LLMs). Benchmarks used to determine processing difficulty have been too English-centric, and heavily biased towards punishing long distances. Our results show that LLMs need to learn how to make use of syntactic signposts in order to process morphologically-rich languages.

Further Research Directions
• Other Languages: Applying the methodology to more varied language families - for example, heavily agglutinative or polysynthetic languages - could yield more compensation strategies besides case-marking.
• LLM Architecture Comparison: Further research could investigate differences in the weights placed on syntactic signposts by other LLM architectures (such as Llama 3 or Mistral) vs. the baseline architecture XGLM.

Limitations of the Study
• Dataset Size: Our analysis is constrained by the relatively limited size of the parsed Universal Dependencies treebanks for some languages. A vaster dataset could provide more robust statistical certainty.
• Language Scope: We analysed exactly 10 languages (5 case-marking and 5 non-case-marking). While illustrative, this is a small typological sample.
• Proxy for Human Difficulty: We utilized LLM surprisal as a proxy for human reading difficulty. While correlated, direct human reading time studies (like eye-tracking) would be needed to absolutely confirm the cognitive reality of these effects.

////Because of space constraint we were not able to add our all graphs in the report So we are providing a drive link :
https://drive.google.com/drive/folders/1mDXGTsk0xy-ahizf65H4SnPEu23ds1Ut?usp=sharing

PART 1 : Emergent Syntactic Efficiency: A Cross-Linguistic Study of Dependency Length Minimization in Grok and Qwen

https://github.com/OM-Nuli/CGS410_course_project.git
https://www.kaggle.com/code/ankit240134/cgs-410 (all json files of our language-data used to generate results are provided in the above github link)

1. Yadav, H., Mittal, S., & Husain, S. (2022). A Reappraisal of Dependency Length Minimization as a Linguistic Universal.
2. Futrell, R., Mahowald, K., & Gibson, E. (2015). Large-scale Evidence of Dependency Length Minimization in Natural Language. Proceedings of the National Academy of Science.

PART 2 : Centre-Embedding and Working Memory in Language Models: A Comprehension and Attention Analysis of GPT-2 and Qwen2.5

The code for centre-embedding is given in the following github link:

https://github.com/Chaitanya197633/CGS-centre_embedding
////The code will take 7-8 minutes to run////

The results we got from the above code are :
results_rows (1).csv.xlsx
https://1drv.ms/w/c/2a2bb9664b0825d2/IQCLaWzs-v4zTrCl5f5EAhlQAeGUxocDTnYqYKUSyMCp9fA?e=HxibRC

1. Elena Voita, et al. (2018). Analysing multi-head self-attention: Specialized heads do the heavy lifting, the rest can be pruned.
2. Soo Hyun Ryu and Richard Lewis (2021). Accounting for agreement phenomena in sentence comprehension with transformer language models.

PART 3 : How Case Marking Helps Languages Handle Long-Distance Word Relationships

https://github.com/OM-Nuli/DLM-case_marking.git

1. Futrell, R., Mahowald, K., & Gibson, E. (2015). "Large-scale evidence of dependency length minimization in 37 languages." PNAS.
2. Levy, R. (2008). "Expectation-based syntactic comprehension." Cognition.
3. Futrell, R., & Levy, R. (2017). "Do people minimize dependency length to help the reader? A computational evaluation." CogSci.